Zhaozheng Shen

zhaozhengcee@u.nus.edu | +65 8965 1656

Education

National University of Singapore Singapore
PhD in Civil and Environmental Engineering *Aug. 2025 – Present*
- Advisor: Assoc. Prof. Justin Yeoh
- Research focus: construction robotics, reinforcement learning, world models, and modular construction.

Tongji University Shanghai, China
Master in Civil Engineering *Sep. 2022 – May 2025*
- Advisor: Prof. Jie Wu
- Thesis: Construction Scheduling for Steel-Frame Structures Using Intelligent Algorithms

Tongji University Shanghai, China
Bachelor in Civil Engineering *Sep. 2018 – Jul. 2022*
- Thesis: Development of a BIM Construction Management Platform for Water Plant Buildings

Research Interests

Construction robotics; reinforcement learning and world models for robot control; simulation-based robot learning; computer vision for construction; modular construction; BIM and digital twins; construction scheduling.

Publications (* corresponding author)

- Jiang, Shanshan; Wang, Yifan; Yang, Bin; **Shen, Zhaozheng**. Reinforcement Learning–Driven Intelligent Path Planning for Dual Tower Cranes in Modular Construction Sites. In *Proc. 43rd International Symposium on Automation and Robotics in Construction (ISARC)*, 2026.
- **Shen, Zhaozheng**; Wu, Jie*. Multiobjective Ant Colony System Algorithm for Component-Level Construction Schedule Optimization. *Journal of Construction Engineering and Management*, 151(3), 04025002, 2025. doi:10.1061/JCEMD4.COENG-15827
- **Shen, Zhaozheng**; Wu, Jie*; Cao, Yilun. A dual-adaptive directed genetic algorithm for construction scheduling. *Journal of Building Engineering*, 96, 110659, 2024. doi:10.1016/j.jobe.2024.110659

Research Experience

Learning-Based Tower Crane Control Mar. 2026 – Present
National University of Singapore, Advisor: Assoc. Prof. Justin Yeoh
- Develop reinforcement-learning controllers for wind-disturbed load transport, active braking, and anti-sway.
- Evaluate payload swing, arrival overshoot, residual motion, and transport time in simulation and Unity.
- Investigate physics-informed controller calibration and continued training from dynamic initial states.

Robotic Interior Finishing for PPVC Modules Aug. 2025 – Present
Research Engineer, National University of Singapore
- Designed robotic interior-finishing concepts for prefabricated bathroom units and developed Isaac Sim simulations of tiling and floor-finishing operations.
- Processed floor point clouds to fit reference planes, measure local slopes, and analyze floor-surface geometry.
- Analyzed construction workflows through site investigations, task-duration measurements, process sequencing, and workflow simulation.

Dual-Tower-Crane Path Planning in Modular Construction Oct. 2024 – Jun. 2025
Tongji University, Shanghai, China
- Contributed to the reinforcement-learning reward design and curriculum training for cooperative dual-crane path planning in a BIM-based Unity simulation (ISARC 2026).

Visual Next-Step Prediction in Robotic Disassembly Jul. 2024 – Present
Research Collaboration, Northeastern University (Remote), Advisor: Prof. Yang Ye
- Annotated over 1,000 robot-vacuum disassembly images and trained YOLOv8 models for component detection.
- Developed RIVET, a multi-frame YOLO–Transformer that encodes detections as ternary component states and predicts the next disassembly action from temporal context (first-author manuscript in preparation).

- Designed robot-wise five-fold cross-validation to test generalization to unseen robot units; the sequence model outperformed a YOLO-CLS baseline in accuracy (0.57 vs. 0.53) and macro-F1 (0.54 vs. 0.50).

Intelligent Construction Scheduling for Steel-Frame Structures Oct. 2022 – May 2025
Master's Thesis Research, Tongji University, Advisor: Prof. Jie Wu

- Parsed IFC geometry and derived constructability constraints through bounding-box contact detection and structural-stability rules, yielding 2,765 precedence constraints for a 1,165-member steel frame.
- Proposed a dual-adaptive directed genetic algorithm (DADGA) with constraint-guided mutation that generates feasible installation sequences more efficiently than a conventional GA (*J. Build. Eng.* 2024).
- Developed a multiobjective ant colony system (MOACS) for multi-skill crew scheduling that balances duration, cost, and carbon emissions, benchmarked against NSGA-II and MOPSO (*ASCE JCEM* 2025).
- Formulated dynamic scheduling under crew absence, late deliveries, and equipment faults as a Markov decision process on disjunctive graphs, using a Dueling Double DQN with prioritized replay to select dispatching rules.
- Implemented the scheduling software and 4D BIM visualization of member-level installation sequences.

BIM Construction Management Platform Development Mar. 2022 – May 2022
Team Member (Bachelor's Thesis Project), Tongji University, Advisor: Prof. Jie Wu

- Parsed geometric location and material information of building components from IFC files.
- Designed Neo4j and MongoDB storage for project files, components, property sets, and their relationships.
- Developed cost, equipment, and document management modules using HTML, CSS, and JavaScript.

Professional Experience

Teaching Assistant, Graduate School Office Sep. 2023 – Feb. 2024
Tongji University, Shanghai, China

- Coordinated teaching arrangements, maintained academic records, and supported graduate thesis defenses.

Teaching Assistant, Engineering Drawing Sep. 2022 – Feb. 2023
Tongji University, Shanghai, China

- Supported 41 students in Descriptive Geometry and Engineering Drawing through grading, tutoring, and exams.

Undergraduate Mentorship Program Apr. 2021 – Sep. 2021
Tongji House Quality Inspection Station, Shanghai, China

- Supported structural inspections and CAD documentation for an 8,000 m^2 paper-mill modernization project.

Construction Management Intern Jul. 2021 – Aug. 2021
Shanghai Mechanized Construction Group Co. Ltd, Shanghai, China

- Assisted with verification of over 1,000 CAD detail drawings, site documentation, and progress tracking.

Awards and Service

- Student Helper, 43rd International Symposium on Automation and Robotics in Construction (ISARC) 2026
- Graduate Student Academic Scholarship, Tongji University 2024, 2023, 2022
- Volunteer, Second National Intelligent Construction Academic Conference Dec. 2023
- Third Prize, Undergraduate Excellent Student Scholarship, Tongji University 2021

Skills

Learning and control: Reinforcement learning, world models, imitation learning; PID, MPC
Simulation and perception: NVIDIA Isaac Sim, Unity 3D, CoppeliaSim; YOLO, Transformers; point clouds
Optimization: Evolutionary algorithms, nonlinear programming, multiobjective optimization
Programming and BIM: Python, MATLAB, JavaScript, Java; Neo4j, MongoDB; IFC, Revit, Tekla, AutoCAD